over-steering an already-conditioned generation step, causing a drop in generation fluency. For this reason, we propose to derive property-dependent α values dynamically at every generation step to intervene with appropriate intensity and 'deactivated' when no longer necessary, limiting the impact of steering on fluency. The key advantage of this strategy is to enable out-of-the-box steering for any property of interest without having to carefully tune the α value beforehand.

6.1 Formulation

Let f be an unsteered LLM and f_Δ be its property-steered counterpart using $\alpha = 2$ for activation injection. For every generation step i, we compute the respective probability distributions over their common vocabulary $\mathcal{V}$ as:

$$p_i^{\varnothing} = \text{softmax}(f(q_{n+1}, y_{<i}))$$
$$p_i^{\Delta} = \text{softmax}(f_\Delta(q_{n+1}, y_{<i}))$$

Intuitively, $p_i^{\varnothing}$ shows the original model predictions, while p_i^{Δ} shows predictions after high-intensity steering is performed. We then compute two vocabulary subsets $Q_i^{\varnothing}, Q_i^{+} \subseteq \mathcal{V}$ by selecting for each of the distributions only the most likely tokens with a cumulative probability of at least p_{top}, as in nucleus sampling[11] (Holtzman et al., 2020):

$$Q_i^{\varnothing} = \{t \in \mathcal{V} \mid \sum_{t_j \leq t} p_i^{\varnothing}(t_j) \leq p_{\text{top}}\}$$
$$Q_i^{+} = \{t \in \mathcal{V} \mid \sum_{t_j \leq t} p_i^{\Delta}(t_j) \leq p_{\text{top}}\}$$

where tokens t_j are sorted in descending order according to respective p_i scores. The union of selected tokens $Q_i = Q_i^{\varnothing} \cup Q_i^{+}$ can be used to filter probability distributions as:

$$\tilde{p}_i^{\varnothing} = \text{softmax}(\{s_j \in f(q_{n+1}, y_{<i}) \; \forall t_j \in Q_i\})$$
$$\tilde{p}_i^{\Delta} = \text{softmax}(\{s_j \in f_\Delta(q_{n+1}, y_{<i}) \; \forall t_j \in Q_i\}) \quad (6)$$

Finally, the α_i value for the selected property corresponding to the current step is computed using the Kullback-Leibler divergence (KL) between the two truncated distributions, bounding the result within the $[0, 2]$ interval to avoid excessive steering:

$$\alpha_i = \min\left(\text{KL}\left(\tilde{p}_i^{\varnothing} \parallel \tilde{p}_i^{\Delta}\right), 2\right)$$

[11] We use $p_{\text{top}} = 0.4$ in Section 7, and include results for $p_{\text{top}} \in [0.4, 0.5, 0.6, 0.7, 0.9]$ in Appendix F

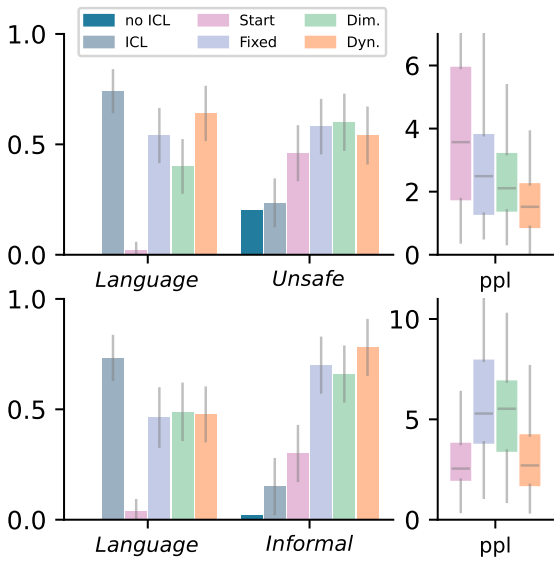

Figure 4: Multi-property steering results for different *languages* (averaged) alongside the *Unsafe* (top) and *Informal* (bottom) properties, respectively. `Dyn` shows the best overall generation fluency while achieving high steering performances.

where KL $\in \mathcal{R}_0^+$. The usage of KL-divergence in this setting is motivated by recent work using similar contrastive metrics to detect context usage in LLM generations (Vamvas and Sennrich, 2021, 2022; Sarti et al., 2024), with the notable difference that in `Dyn` the shift in probabilities is produced by activation steering rather than additional input context. Intuitively, this method allows for modulating steering intensity at every step i according to the expected shift produced by high-intensity steering ($\alpha = 2$). If steering would not produce a significant shift in probabilities due to an already-conditioned prefix $y_{<i}$ for step i, the resulting $\alpha \simeq 0$, avoiding over-steering and preserving model fluency whenever possible.

7 Multi-property steering

Under the assumption of linearity of the model's internal activations (see Section 2), we evaluate baseline activation injection strategies and the newly introduced `Dyn` method for multi-property steering, focusing in particular on conditioning model outputs to match the *Unsafe* or *Informal* properties while also requiring them to be in one of the four studied languages. All activation injection techniques (`Start`, `Fixed`, and `Dim`) and the `ICL` and `noICL` baselines tested in Section 5 are evaluated alongside Dynamic Activation Composition (`Dyn`).

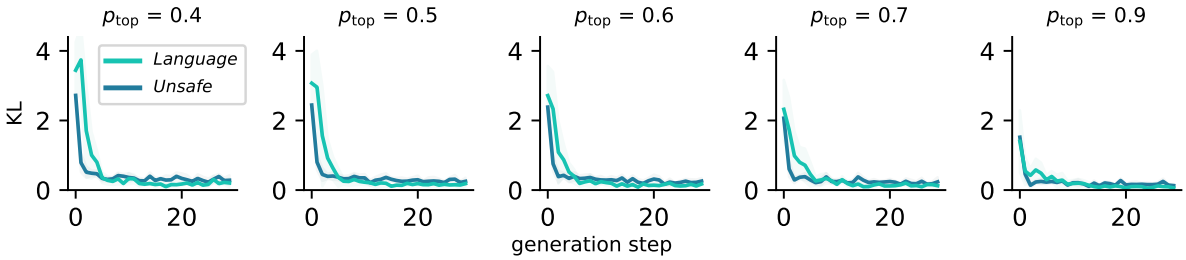

Figure 5: Avg. α_i scores produced by the Dyn method for multi-property steering of Unsafe and Language properties using $p_{\text{top}} \in [0.4, 0.5, 0.6, 0.8, 0.9]$. Overall, stronger steering intensity is only required on the first few generated tokens. Exceptions to this behavior are discussed in Appendix G.

Results Figure 4 shows multi-property steering results for different conditioning techniques averaged across all available languages.[12] In most cases, Dyn yields the best trade-off between steering strength (higher accuracy) for each task and generation quality (lower $\Delta\text{PPL}_{\text{ICL}}$). We note in particular how, in multi-property settings, language conditioning dominates the result in the ICL case, while the *Unsafe* and *Informal* aspects in the provided examples are mostly ignored by the model. In contrast, the various injection strategies achieve good conditioning on both properties with minimal increases in perplexity.

By examining the steering intensity applied in the Dyn setting during generation (Figure 5 shows an example for *Language* (averaged) and *Unsafe*), we note that α_i generally decreases sharply after the first few generated tokens, suggesting that our naive Dim strategy might still overestimate α_i values at intermediate generation steps. Generation examples in the Dyn setting[13], show that the language steering intensity decreases as soon as a few complete words in the desired *language* are generated. Similarly, for *Unsafe* the α value drops as soon as the model generates a sequence of tokens that complies with the prompt's unsafe request.

Lastly, Figure 5 also shows that the p_{top} parameter, which determines the amount of tokens considered in the KL Divergence computation, shows a negative correlation with the sharpness of the initial spike in α values: the smaller the value, the more restrictive the top-p token selection, leading to a higher KL. Intuitively, for higher values of p_{top} many of the selected tokens would receive negligible probability mass from both the steered and the unsteered model, leading to an under-estimate of the steering required. Across all tasks, we find 0.5 as the optimal value for p_{top}, leading to a sufficiently low cardinality of Q to capture probability shifts between most likely tokens that could be selected by sampling or beam-search decoding.

8 Conclusion and future work

Through a systematic study of different activation injection strategies, we confirm that activation steering is an efficient and promising way to condition LLM generations on desired properties. However, we also observe that existing injection techniques are limited in two ways: (i) steering beyond single tokens, i.e., ensuring that the conditioning is preserved across longer generations, requires interventions that harm output fluency; (ii) their effectiveness is property-dependent, making it challenging to steer multiple properties simultaneously as each property is likely to require an ad-hoc steering intensity to ensure maximal performance. For this reason, we proposed Dynamic Activation Composition, a strategy to adaptively control the steering intensity at each generation step according to the expected steering effect, thereby limiting over-steering of already-conditioned properties while promoting the under-conditioned ones, ultimately achieving the best trade-off between conditioning accuracy and output fluency.

In sum, Dynamic Activation Composition can facilitate the alignment of LLMs to multiple desired properties and behaviors at once. In future experiments, it will be interesting to study the effect of our method on the perplexity of larger LLMs, considering these models are naturally more fluent. From an interpretability standpoint, our approach offers an interesting direction to study how properties condition model behavior during generation.

[12] The best α configuration is selected for each technique, i.e. $\alpha_{Language} = 1, \alpha_{Unsafe} = 1.5, \alpha_{Informal} = 1$, and $p_{\text{top}} = 0.4$ for Dyn. Full results in Appendix F.

[13] Examples available in Appendix D and G

Limitations

The advantage of Dynamic Activation Composition is evident from the comparison to the other techniques that we test. However, the results we report are based on experiments with one instruction-based model only, namely Mistral 7B. A more comprehensive study should include a larger range of models, both in terms of size and characteristics, for example whether they have been instruction-tuned or aligned via RLFH.

In order to obtain the manual composition for *Language* and *Unsafe/Informal* we use machine-translated datasets, either existing ones, such as Alpaca, or specifically created in the context of this study. While this is common practice, and manual inspection has revealed a high quality of the translations, optimally one would use, especially for the *Language* steering, original texts exhibiting the properties of interest in the chosen languages.

For evaluating the outputs, we use previously developed, high-accuracy models and perplexity. A larger-scale experimental setup could also include human judgments over generations to ensure the reliability of those metrics.

Finally, we limit our evaluation of injection strategies to a single steering setup (described in Section 3), which is in line with previous work using contrastive pairs of in-context examples for activation steering. Future work could evaluate whether our proposed `Dyn` method would generalize to other steering configurations using, for example, the directions derived from probing classifiers.

Ethics Statement

While this work's core contribution is technical in nature, we are aware that the Dynamic Activation Composition technique that we propose can, in principle, be used with malicious intents aimed at amplifying potentially harmful model behavior. However, techniques like Dynamic Activation Composition allowing for a deeper intervention on the model's behavior might prove more comprehensive, controllable and robust than RLHF in the future. Hence, we believe that the relevance of this research outruns concerns due to dual use-associated risks. More in general, in spite of potential misuses, we do believe in the importance for the research community of maintaining a line of work focused on enhancing the adaptability and transparency of models' behaviors.

Acknowledgements

The work of Daniel Scalena has been partially funded by MUR under the grant ReGAInS, *Dipartimenti di Eccellenza 2023-2027* of the Department of Informatics, Systems and Communication at the University of Milano-Bicocca. Gabriele Sarti acknowledges the support of the Dutch Research Council (NWO) as part of the project InDeep (NWA.1292.19.399). We also thank the Center for Information Technology of the University of Groningen for providing access to the Hábrók high-performance computing cluster used in fine-tuning and evaluation experiments.

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